



Validating a measure of growing pattern understanding in Chinese preschool children

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ABSTRACT

Recent research has pointed to the importance of patterning skills in early mathematics development. However, there is a lack of psychometrically robust instruments to measure patterning skills, especially growing pattern understanding. In this study, we developed and validated an instrument assessing growing pattern understanding among preschool children. One hundred and thirty-eight preschoolers aged approximately five years in Hong Kong were assessed individually on 10 growing pattern items at two time points over a one-year period. Confirmatory factor analysis showed that a two-factor model (i.e., increasing and decreasing patterns) fit the data best in the second year of preschool (PS2), but a one-factor model (i.e., growing patterns) fit the data best in the third year (PS3). Scale reliability was adequate at both time points. At the item level, Rasch analysis showed that all the items were within an acceptable range of difficulty at both time points. Information curves also suggested that the test items provided the most information (i.e., having high reliability) at high ability levels at PS2 but were a sensitive measure for children of both low and high abilities at PS3. No differential item functioning was detected across boys and girls at PS2 or PS3. Also, expected relations were found between growing pattern understanding and children's performance on repeating patterns and English expressive vocabulary, supporting convergent and discriminant validity. Overall, the results suggest that the growing pattern task has sound psychometric properties and is appropriate for use in preschool children aged five and six years.

1. Introduction

In recent years, there has been increasing recognition in the field of mathematics cognition and learning that successful mathematics learning depends not only on numerical skills, but also on non-numerical skills (Borriello et al., 2022; Rittle-Johnson et al., 2019). One of such skills that has generated intense research interest is pattern understanding, also known as patterning ability, patterning skill, or simply patterning (Wijns et al., 2019). Pattern understanding refers to the ability to detect regularities or patterns in a predictable set of informational units (e.g., ABAB or 1-3-5-7; Rittle-Johnson et al., 2015), be that information spatial, numeric, or verbal. The importance of pattern understanding in mathematics education has been supported by a number of empirical studies, which showed that children's performance on patterning tasks is linked to a wide range of mathematics learning outcomes, such as arithmetic (Burgoyne et al., 2019), calculation (Borriello et al., 2022), algebra (Wilkie & Clarke, 2016), and fraction knowledge (Mielicki et al., 2021).

Although there exist many types of patterns, research on preschool

children has focused predominantly on simple repeating patterns, which are linear patterns that feature a constant repeating unit (e.g., ABAB). Growing patterns, which involve a systematic increase or decrease in the underlying pattern rule (e.g., ABAABAAAB or 30-20-10), have received limited attention (Burgoyne et al., 2019; Wijns et al., 2021; Zhang et al., 2017). The scant attention to growing patterns is primarily due to the significant difficulty many four- and five-year-olds experience in understanding these pattern problems (Gadzichowski, 2012), which fuels the argument that they may not be appropriate for preschool children (Pasnak et al., 2019). However, studies have shown that some children in this age group are already capable of solving growing pattern problems (e.g., Mulligan et al., 2020; Wijns et al., 2019) and performance on these tasks can predict later mathematics skills over and above repeating patterns (e.g., Borriello et al., 2022). Taken together, these studies suggest that greater attention needs to be given to growing patterns for young children in early mathematics learning and instruction.

One impediment to further research on growing patterns, however, is the lack of psychometrically sound instruments. Only a couple of measures assessing growing pattern understanding can be identified in the

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literature, and existing studies often do not report any validity evidence (e.g., dimensionality or criterion-related validity) beyond Cronbach's α . The lack of the necessary psychometric groundwork may undermine the quality of inferences drawn from these studies. Therefore, the goal of the present study was to develop and validate a measure of growing pattern understanding in preschool children and provide evidence supporting its use for future research.

1.1. Growing Patterns

Despite the potential importance of young children's growing pattern understanding, several studies have shown that four- to five-year-old children tend to struggle with growing pattern problems and many fail to grasp the pattern rule (e.g., Papic et al., 2011; Wijns et al., 2019). To understand young children's struggle with growing patterns, several explanations have been proposed. One explanation holds that growing patterns may involve high cognitive demands as the core elements in a growing pattern change systematically along one or more dimensions (Mielicki et al., 2021; Warren & Cooper, 2008). To identify the pattern rule, multiple consecutive elements need to be compared, which likely requires great working memory resources (Borriello et al., 2022) and may not be fully developed in preschool children. This view is also echoed by Papic et al. (2011), who note that growing patterns may involve functional thinking as multiple relations need to be processed when children solve growing pattern problems (e.g., relations among the elements in a sequence as well as their position in the sequence). Another explanation attributes the difficulty of growing patterns to a lack of targeted training in educational practice. Since traditional patterning instruction and practice places heavy emphasis on repeating patterns, especially simple alternating patterns such as ABAB (Papic et al., 2011; Warren, 2005; Wijns et al., 2019), children have few opportunities to get familiar with other types of patterns such as growing patterns. As a result, children may resort to the same strategies they use in solving repeating pattern problems when they come across growing patterns. Such a skewed understanding likely contributes to children's incorrect responses to growing pattern tasks (Papic et al., 2011; Warren, 2005).

Although growing patterns tend to be difficult for preschoolers, not all children fail to perform successfully on growing pattern tasks. In a study investigating the patterning skills of four-year-olds, Wijns et al. (2019) found that while the growing pattern task was difficult for the participating children, the growing pattern activities were "feasible for a number of children of this young age" (p. 159). Besides, although some task formats such as number series (e.g., "1, 3, 5") may be excessively challenging for preschoolers, task complexity can be simplified by using non-numerical growing patterns (Jordan et al., 2006; McKillip, 1970; Stephens et al., 2017; Wijns et al., 2019). In addition, Papic et al. (2011) showed that while many children aged three to five years initially failed the growing pattern tasks, most children were capable of extending and explaining growing patterns after one year of training. These studies provide evidence for the feasibility of assessing growing pattern understanding as early as the preschool years and also point to the promise of relevant interventions.

Understanding growing patterns in the early years is important because it may serve as a unique pathway to mathematics learning. McKillip (1970) argued that non-numerical growing patterns could help children establish connections between patterning and numerical skills such as "one more than." Understanding and generalizing the rule underlying growing patterns could also lay the foundation for algebraic or functional thinking in later grades (e.g., Warren & Cooper, 2008). Correlational studies have provided evidence that young children's performance on growing pattern tasks predicts their conceptual and procedural mathematics knowledge (Borriello et al., 2022). These effects persisted even after repeating patterns and other cognitive skills were controlled for. Besides, the predictive effect of growing patterns was only observed for children who were approximately five years old,

but not for older children (i.e., up until the age of nine; Borriello et al., 2022). Together, these studies highlight the importance of studying growing pattern understanding for young children in the preschool period.

1.2. The Measurement of Growing Pattern Understanding

Despite a surge of interest in growing pattern understanding, few instruments are available to measure this construct for young children (e.g., Borriello et al., 2022; Mielicki et al., 2021; Wijns et al., 2019; Wijns et al., 2021). Growing patterns can be assessed using a variety of formats (e.g., letters, objects, geometric figures, number sequences, tables of input and output values, etc.). For preschool-aged children, the extant instruments mostly use physical objects or visual figures (e.g., Borriello et al., 2022; Mielicki et al., 2021). Using concrete objects and figural materials can help children establish a connection between position number and the number of units in a growing pattern. Presenting growing patterns in the form of figures, compared with numerical sequences, also led to more variability in children's patterning performance and resulted in better performance on far extension tasks (Mielicki et al., 2021). There is also preliminary evidence that children as young as five years old can generalize growing patterns presented in the form of figures (e.g., Stephens et al., 2017).

A variety of activities can be used to measure growing pattern understanding, including copying or duplication, extending, translating or generalizing, and pattern identification (Borriello et al., 2022; Wijns et al., 2019; Wijns et al., 2019). Copying requires children to make a replica of a pattern using the same materials. Extension requires children to continue the pattern. Translating requires children to make the same pattern with different materials (also termed abstracting; Wijns et al., 2019), whereas pattern identification requires children to look at a pattern and then identify it after it disappears (Borriello et al., 2022). The latter two activities are generally more difficult as they require insight into the underlying rule (Mulligan et al., 2006; Wijns et al., 2019).

There are a few problems with the existing instruments. First, the operationalization of growing pattern understanding is often ill-matched with its conceptual definition. While growing patterns are often theorized to involve both increasing and decreasing elements (e.g., ABAABAAAB, 30-20-10; Mielicki et al., 2021), many tasks that were designed to tap growing pattern understanding consist of increasing patterns only (e.g., Borriello et al., 2022). Such an instrument is not aligned with the conceptual definition and may not be representative of the target domain. Therefore, it may not capture all aspects of this construct and lack content and construct validity. Second, a single sum score was usually calculated to represent growing pattern understanding even when both increasing and decreasing patterns were assessed (e.g., Jordan et al., 2006). However, without sufficient psychometric evidence on the dimensionality of this construct, it is unclear whether increasing and decreasing patterns represent distinguishable components or inconsequential task attributes, and using a single score may not be fully justified. Moreover, the psychometric evidence for most existing instruments is based on cross-sectional data, which can only provide a snapshot of an instrument's performance at a single point in time. Whether the same instrument has satisfactory psychometric properties at different time points remains unknown. Without such evidence, the use of a certain instrument for studying children's development may be questioned. To address the above gaps, the measure we developed in this study included both increasing and decreasing patterns and was validated based on longitudinal data of a group of children who were followed over a one-year period.

1.3. The Present Study

Due to the lack of available instruments for assessing growing pattern understanding in preschool settings, the goal of this study was to develop an instrument assessing growing pattern understanding and gather relevant validity evidence that supports its use among preschool

children. Specifically, we developed a measure of growing pattern understanding that was aligned with its conceptual definition that includes both increasing and decreasing types of pattern items. It required children to extend patterns as this activity was appropriate for studying growing patterns in preschool-aged children (e.g., Mulligan et al., 2006; Wijns et al., 2019). Based on relevant research and in line with current assessment practices with preschool children (Mielicki et al., 2021; Stephens et al., 2017; Wijns et al., 2019), we also used geometric figures and familiar objects as test materials rather than numbers. Moreover, we used confirmatory factor analysis and Rasch measurement to validate the factor structure and psychometric properties of this instrument (i.e., item difficulty and internal consistency). In addition, we investigated differential item functioning across sex to see whether the test items were fair to boys and girls. This is because fairness is increasingly emphasized in the development and validation of tests or assessments, as is recommended in professional documents such as the *Standards for Educational and Psychological Testing* (American Educational Research Association [AERA] et al., 2014). Establishing the equal functionality of a test can help ensure the proper interpretation and use of test scores for different groups. Finally, we established convergent and discriminant validity by correlating growing patterns with criterion variables. We hypothesize that the measure we developed in this study would have sound psychometric properties and be developmentally appropriate for use in preschool-aged children.

2. Method

2.1. Participants and procedures

Altogether, 138 children aged approximately five years ($M = 4.98$; $SD = .33$) were recruited from four preschools in Hong Kong to participate in this study. Cantonese was the primary language used in these kindergartens and all children were fluent Cantonese speakers. Although we did not collect ethnic information, field observations showed that the children were mostly of Chinese descent (i.e., mainland China and Hong Kong), with ethnic minorities constituting a very small minority, which is consistent with the general population composition in Hong Kong (Census and Statistics Department, 2016). The sample had a balanced gender ratio, with 49.28 % of the children being girls ($N = 68$). The children were from middle- to upper-class families, with 53 % of households having a monthly income of HK\$40,000 or more (about US \$5103.60 in 2018). Regarding educational background, of the 100 (72.5 %) fathers and 108 (72.5 %) mothers who reported demographic information, 39 % of the fathers and 36 % of the mothers had a bachelor's degree or above.

In Hong Kong, most children receive three years of preschool education from ages 3 to 6 before they enter primary school. Unlike in some western countries, the terms “kindergarten” and “preschool” are used interchangeably in Hong Kong. We have opted to use the term “preschool” consistently throughout the paper to ensure clarity and maintain consistency. In most Hong Kong preschools, including those included in the current study, Cantonese is used as the primary language of instruction, while English and Mandarin are used as supplementary languages (Zou et al., 2022).

Before starting our project, we submitted our project application materials, including research proposal, test materials, and consent forms, to the Institutional Review Board at the corresponding author's university. After a thorough and rigorous review, the board approved our study. Then, we contacted and obtained consent from the principals of some local preschools. Afterwards, we asked the teachers to distribute a package to parents, which included the consent form and questionnaires. Parents who agreed to participate were then recruited for our study. Before the formal testing, well-trained testers, who were undergraduate or master's students, participated in a training session. The training was conducted by the principal investigator (PI) of the project and a research assistant (RA) and lasted approximately 3 hours. The training session comprised two parts. Part one involved

familiarization with the content of the assessments. Each tester was provided with a test booklet and walked through the specifics of the assessments, such as how to give instructions, ask questions, and accurately score children's responses. Instruction was also given on how the testers could build rapport and interact appropriately with children. In part two, the testers formed pairs with each other or with the PI or RA and engaged in a mock testing session, taking turns as both the tester and the child. Only those who were considered to have followed the standard procedures and conducted the assessments appropriately successfully completed the training. Throughout the training, clear guidance and standardized guidelines were provided to ensure consistency across testers, but no formal calculation of interrater reliability was performed. To address any questions and confusion that might arise, a Q&A session was also arranged to answer the testers' questions about any aspect of the assessments, further solidifying their understanding and preparing them for the formal testing.

Then, each child was individually assessed in the spring semester of their second preschool year (PS2; Time 1 [T1]; between May and June of 2018) at the first wave of assessment. The second wave took place one year later (PS3; Time 2 [T2]). The number of children who dropped out was small ($N = 12$) and dropout was due to children transferring to another preschool. Attrition analysis showed that the dropout of participants at T2 was not related to children's responses to the growing patterns items, their age, sex, family income, father's education, or mother's education at T1. At each wave, the assessment was conducted in Cantonese by the well-trained testers in a quiet room in the children's respective schools. The assessment took about one hour for each child, and it was broken down into three to four sessions to avoid children's fatigue. After completing the assessment, the children received a thank-you gift such as a pencil case for their participation.

2.2. Measures

2.2.1. Growing patterns. Children's understanding of growing patterns was assessed at T1 and T2 using a task that was designed by the corresponding author (see online Appendix A). The test consisted of five increasing patterns and five decreasing patterns. Children were invited to play a game using pictures featuring an increasing or decreasing pattern and were instructed to arrange the pictures according to the pattern. The task materials were familiar objects in children's daily life, such as ice-sugar gourds, building blocks, sticks, and radishes. In the increasing pattern items, children were given a series of pictures by the trained testers and were asked to answer what the next picture would be. For example, the testers placed four ice-sugar gourd sticks in front of the child and placed one, two, and three ice-sugar gourds on the first, second, and third sticks, respectively. The testers then asked the child to follow the pattern rule and decide how many ice-sugar gourds to place on the fourth stick (i.e., four). Similarly, in the decreasing pattern items, children were asked to arrange the subsequent picture following a series of pictures featuring a decreasing pattern. For example, the testers presented caterpillars as circles in front of the child, with the caterpillars in the first, second, and third rows containing 12, 9, and 6 circles, respectively. The child was then asked to follow the pattern rule and arrange a certain number of circles for the caterpillar in the fourth row in the right pattern. A total of 10 items were administered and one point was given for each correct response, with a maximum score of 10. For illustration, two example items are shown in Fig. 1.

2.2.2. Repeating patterns. We translated Rittle-Johnson et al.'s (2015) repeating pattern task and used it at T1. For this task, children were shown a series of two-dimensional and three-dimensional geometric shapes that involved the repetition of a certain shape or color sequence. Then, they were asked to perform various activities, such as duplicating or extending the pattern, abstracting it, memorizing it, or identifying the pattern unit. To give an example of the extending activity, a child was



Fig. 1. Example Items of an Increasing Pattern and a Decreasing Pattern.

presented with six shapes that featured one rhombus followed by two triangles and then asked to extend the pattern by arranging what shapes should follow from this sequence. If the child correctly placed one or more combinations of one rhombus followed by two triangles, one score was given. A total of 10 items were used. Ordinal α was .89 at T1.

2.2.3. English expressive vocabulary. This was measured using a picture naming task (Learning Disabilities Association of Alberta, 2009). This measure has been successfully used for children of similar ages in Hong Kong (e.g., Yeung & Chan, 2013; Yeung et al., 2013) and has demonstrated good reliability. Children were presented with a series of pictures featuring a certain object, animal, or person and asked to name the entity in English. Target concepts that were assessed in this measure include “sun”, “boy”, “frog”, “fish”, “chair”, “snake”, “pear”, among others. Testing proceeded to the next trial when a child spent more than five seconds on each item. The testing was terminated when a child made five consecutive mistakes. A total of 15 items were used and one score was given for each correct response. Ordinal α was .97 at T1.

2.3. Analytic strategy

A four-step analytic strategy was adopted. In the first step, we performed confirmatory factor analysis (CFA) to validate the factorial structure of the growing patterns test at T1 and T2. Both one-factor and two-factor models were specified. The one-factor model corresponded to the hypothesis that the growing patterns test measured a single underlying dimension (i.e., growing patterns), whereas the two-factor model corresponded to the hypothesis that the test was comprised of two different but interrelated dimensions (i.e., increasing and decreasing patterns). For the two-factor model, the first five items (i.e., Items 1-5) were specified to load on increasing patterns while the last five items (i.e., Items 6-10) loaded on decreasing patterns. The weighted least squares estimator with means and variance adjustments (WLSMV) for ordered categorical data was chosen since all the test items were dichotomously scored. To evaluate model fit, the following fit statistics were used: the chi-square statistic (χ^2), the goodness of fit index (GFI; $\geq .90$), comparative fit index (CFI; $\geq .95$), Tucker-Lewis index (TLI; $\geq .95$), and root mean square error of approximation (RMSEA; $\leq .06$) (Hu & Bentler, 1999). Although the standardized root mean squared residual (SRMR) is routinely reported in the context of CFA analysis with continuous data, Yu (2002) recommended against the use of SRMR with binary outcomes based on results from simulation studies. Therefore, this statistic was not used for model fit evaluation in this study. For model comparisons, the chi-square difference test was performed and changes in CFI values (Δ CFI) were also reported. A nonsignificant chi-square test statistic and Δ CFI values less than .01 suggest that the more complex model does not fit significantly better than the more parsimonious model, indicating that the more parsimonious model should be chosen.

In the second step, Rasch modeling was used for item calibration at T1 and T2. The Rasch model is a probabilistic model that links observed item responses to latent abilities or traits. When evaluating the qualities of items, it provides only an item difficulty parameter since it rests on the assumption that a person's response to an item is a function of that person's ability and the difficulty of the item (Bond et al., 2021). In this study, the difficulty of individual test items was inspected to identify

whether they were appropriate for preschool children. In practice, difficulty values within the range of -3 to 3 are generally considered acceptable. Items are of medium difficulty when difficulty values approach 0, difficult when they are greater than 1.50, and easy when they are less than -1.50 (Baker & Kim, 2017). Since person ability and item difficulty are put on the same scale in the Rasch framework, they can be directly compared. For example, a child with an ability score of 2.00 is expected to have a high probability of solving an item with 1.50 difficulty or below and is thus considered a high-ability child. Furthermore, item and test information, which can be interpreted similarly as traditional indices of reliability (e.g., Cronbach's α), was also calculated to display the precision of the test in measuring children's growing pattern skills along the underlying ability continuum. Higher information indicates greater person score estimation precision at the corresponding location of the underlying ability continuum and test information is simply the sum of item information values (de Ayala, 2022). In practice, the focus is usually on the shape of the information functions and the point where information reaches its peak, but the specific information values are largely ignored (Boone & Staver, 2020).

Prior to conducting Rasch analysis, certain requirements need to be met. First, two key assumptions need to be checked, namely, unidimensionality and local independence. Unidimensionality means that only one dimension underlies the test or item responses, while local independence means that after overall differences in ability are controlled, individual responses to any two items are statistically independent (Bond et al., 2021). The former was established based on CFA analysis in the first step, while the latter was examined using Yen's Q_3 statistic (Yen, 1993), which is a measure of residual correlations among item pairs. In the literature, critical values of Yen's Q_3 smaller than .2 or .3 have been suggested as indicating local independence (Chen & Thissen, 1997; Ramp et al., 2009). Second, model-data fit for items was also routinely examined in Rasch analysis with infit and outfit statistics. They serve as indices that gauge the degree of sensitivity to responses to test items that are close to or far away from a person's ability, respectively (Linacre, 2003). For both statistics, usually two sets of values are reported: mean square values (MNSQ) and standardized values (ZSTD). The former quantifies an item's contribution to the productivity of measurement and is calculated as chi-square statistics divided by their corresponding degrees of freedom. Values within the range of .5 to 1.5 suggest that an item is productive for measurement, while values smaller than .5 indicate an item is less productive but not degrading (Linacre, 2002). The latter, namely, the ZSTD statistics, are standardized values that measure the improbability of data. Values within the range of -1.9 to 1.9 suggest reasonable predictability, while values smaller than -2 suggest that data are too predictable and values greater than 2 signify that data are too unpredictable (Linacre, 2002). Finally, item reliability and separation indices were also reported. Rasch-based item reliability can be interpreted similarly as traditional reliability indices such as Cronbach's α , with the ranges for acceptable, good, and excellent reliability being .70-.79, .80-.89, and $\geq .90$, respectively (Duncan et al., 2003). Item separation provides yet another measure of reliability and serves to verify the hierarchy of item difficulty. High separation indicates that a person sample is able to separate the different levels of item difficulty of the instrument. In other words, high separation implies that there is sufficient variation in the difficulty of the items and

redundancy is limited, thereby allowing better differentiation of persons of different abilities. The rules of thumb for acceptable, good, and excellent separation are 1.50–1.99, 2.00–2.99, and ≥ 3.00 , respectively (Duncan et al., 2003).

In the third step, sex differential item functioning (DIF) was investigated at both time points to identify whether any items functioned differently between boys and girls. The logic of DIF is that test takers at the same ability level, regardless of their group membership, should have the same probability of correctly answering an item conditional on differences in their overall ability. Otherwise, the item may have tapped into construct-irrelevant dimensions that unduly favor or disfavor a certain group of respondents (Holland & Wainer, 2012). Overall ability is often represented by a matching score, such as a sum score of item responses. However, when individual items are not DIF-free, the matching score may be contaminated. Therefore, only items that are free of DIF should be used for matching and anchor purification (i.e., iteratively removing DIF items until no additional DIF items are left) is recommended before running a DIF analysis (Pae & Park, 2006). In this study, the classical Mantel-Haenszel procedure with anchor purification was adopted. Throughout the analyses, boys were used as the reference group and girls were used as the focal group (i.e., boys = 0; girls = 1). Adjustments for multiple comparisons were based on the Holm method, which has been shown to perform well for DIF purposes (Kim & Oshima, 2013). A statistically significant result based on the adjusted p values would indicate the presence of DIF in a test item. A positive and negative Mantel-Haenszel statistic for an item indicates that this item favors girls and boys, respectively.

In the fourth step, convergent and discriminant validity of the test were separately established at T1 and T2 by correlating children's performance on the growing patterns task and a set of criterion measures (i.e., repeating patterns and English expressive vocabulary). Since repeating and growing patterns assess different aspects of patterning skills and involve similar cognitive processes (e.g., identifying the regularity), we hypothesized that children's scores on the growing pattern test would be positively correlated with their performance on the repeating pattern task. The choice of a measure for assessing discriminant validity was less straightforward, as previous research has shown correlations between different types of patterning and reading as well as arithmetic skills (e.g., Borriello et al., 2022; Burgoyne et al., 2017; Burgoyne et al., 2019). Since little research has shown a correlation between growing patterns and vocabulary measures, much less in the context of second or foreign language learning, we chose English expressive vocabulary, which should exhibit sufficient distinctiveness to allow us to test discriminant validity. Therefore, we expected that there would be a null or weak relationship between the two.

Regarding missing data, we performed Rubin's MCAR test for the item responses to the growing patterns test at T1 and T2, which both yielded a nonsignificant result (T1: $\chi^2 = 42.20$, $df = 49$, $p = .74$; T2: $\chi^2 = 26.80$, $df = 35$, $p = .84$), suggesting that missing data were not a problem. We used the default missing data handling method in the *lavaan* package (i.e., listwise deletion; Rosseel, 2012) for the main CFA analyses. But considering the impact such deletion might have on the results in light of our relatively small sample, we also performed multiple imputation using the *mice* package (van Buuren & Groothuis-Oudshoorn, 2011) to examine the robustness of our results under different missing data handling methods (Schafer & Olsen, 1998; see Appendix B Supplementary materials). For the Rasch analysis, no deletion was performed and all the available information was taken into account, including cases with missing data, by the *ltm* package (Rizopoulos, 2006). All data analyses were performed in R (R Core Team, 2022), and besides the packages already mentioned, other packages used include *autoRasch* (Wijayanto et al., 2023), *diffR* (Magis et al., 2010), *nanian* (Tierney et al., 2021), and *subscore* (Dai et al., 2022).

3. Results

3.1. CFA analysis

Table 1 summarizes the fit statistics of the one-factor and two-factor CFA models (also see Appendix B Supplementary materials for the CFA analyses with multiply imputed data; the results were consistent with those reported here). At T1, both the one-factor and two-factor models showed satisfactory fit to the data, as shown by the chi-square test result and the comparative fit indices. Furthermore, the chi-square difference test and ΔCFI showed that the two-factor model fit significantly better than the one-factor model, $\Delta\chi^2(1) = 4.40$, $p = .04$; $\Delta CFI > .01$. Therefore, the two-factor model was accepted as the final model at T1. The standardized factor loadings of increasing patterns ranged from .52 to .79, while the loadings of decreasing patterns ranged from .51 to .87 (see Table 2). All loadings were statistically significant. Notably, the two dimensions of this model were empirically distinguishable but highly correlated ($r = .77$). Ordinal coefficient alpha was calculated to estimate the reliability of the two dimensions, which were .86 for increasing patterns and .77 for decreasing patterns, respectively. The reliability of the entire test was also high at .86. Paired samples t -test showed that the children performed significantly better on the increasing patterns dimension ($M = 1.38$, $SD = 1.15$) than on the decreasing patterns dimension ($M = .72$, $SD = 1.00$), $t(137) = 6.82$, $p < .001$, Cohen's $d = .58$, 95 % CI [.40, .76].

At T2, both the one-factor and two-factor models fit the data well. However, the chi-square difference test and the change in CFI showed that the two-factor model did not fit significantly better than the one-factor model, $\Delta\chi^2(1) = 2.22$, $p = .14$; $\Delta CFI < .01$. Besides, the two dimensions of increasing and decreasing patterns were so highly correlated ($r = .87$) that they were empirically indistinguishable. This indicates that the growing pattern test was essentially unidimensional at T2. Therefore, the one-factor model was accepted as the final model at T2. The standardized factor loadings of the test items ranged from .54 to .92 and all were statistically significant (see Table 2). Ordinal coefficient alpha for the entire test was .90.

3.2. Rasch modeling

The simple Rasch model was used to calibrate item difficulty parameters (denoted as δ) for the growing pattern test. Since a two-factor model was chosen for T1 and a one-factor model was chosen for T2, the Rasch model was applied separately to the items assessing increasing and decreasing patterns at T1 but simultaneously to all test items at T2 (i.e., unidimensionality). Yen's Q_3 statistics showed that the residual correlations for all item pairs were smaller than .3 at T1, although for

Table 1
Fit Indices of the One-factor and Two-factor CFA Models at T1 and T2.

Model	χ^2	df	p	CFI	GFI	TLI	RMSEA [90 % CI]
T1							
One-factor model	38.70	35	.31	.99	.96	.99	.03 [.00, .07]
Two-factor model	33.27	34	.50	1.00	.96	1.00	.00 [.00, .06]
Model comparison	$\Delta\chi^2$	Δdf	p	ΔCFI			
	4.40	1	.04	.012*			
T2							
One-factor model	21.02	35	.97	1.00	.98	1.03	.00 [.00, .00]
Two-factor model	18.43	34	.99	1.00	.98	1.03	.00 [.00, .00]
Model comparison	$\Delta\chi^2$	Δdf	p	ΔCFI			
	2.22	1	.14	.000			

Note. T1 $N = 124$. T2 $N = 120$. * $\Delta CFI > .01$; the calculation may not exactly equal the difference between the CFI values reported in the table due to rounding error. One-factor model: growing patterns. Two-factor model: increasing and decreasing patterns.

Table 2
Standardized Factor Loadings of the CFA Models at T1 and T2.

Item	T1		T2	
	One-factor	Two-factor	One-factor	Two-factor
1	.61	.67	.53	.56
2	.68	.73	.63	.66
3	.74	.79	.64	.68
4	.49	.52	.65	.69
5	.72	.77	.61	.65
6	.78	.82	.69	.70
7	.68	.71	.79	.80
8	.48	.51	.69	.70
9	.82	.87	.93	.95
10	.62	.64	.70	.72

Note. T1 $N = 124$. T2 $N = 120$. One-factor model: growing patterns. Two-factor model: increasing and decreasing patterns.

one item pair the correlation was greater than .2 (i.e., Item 1 and Item 2). At T2, the residual correlations were smaller than .2 for all item pairs. These results indicated that the local independence assumption was reasonably satisfied for our analysis.

The infit and outfit statistics as well as item reliability and separation for all the items are reported in Table 3. At T1, nine of the 10 items showed infit MNSQ values within the range of .5 to 1.5. However, the outfit MNSQ values of seven items were smaller than .5, suggesting that they did not contribute much to productive measurement, although they were not degrading. Several items showed ZSTD values smaller than -1.9, indicating that they were too predictable. At T2, infit and outfit statistics were acceptable for nine of the 10 items, with only one item showing some slight deviation (i.e., Item 9). This suggests that although some items did not show good fit at T1, they were still useful measures and could be retained for use at a later time point. Regarding item reliability and separation, both values were greater than the recommended thresholds for excellent reliability and separation across the two time points.

The item difficulty estimates along with their standard errors (denoted as s_e) are reported in Table 4. As shown, all the test items were within an acceptable range of difficulty (i.e., -3 to 3) for the preschool-aged children across the two time points. At T1, the test was relatively difficult for the children, with all but one item (i.e., Item 1) taking on positive difficulty values and five items having values greater than 1.5 (i.e., Item 3, 4, 6, 8, and 10). Among them, Item 1 was the easiest ($\delta = -1.04$) and Item 10 was the most difficult ($\delta = 2.73$). At T2, all the items became easier for these children, as can be seen by the smaller difficulty parameters. None of the items had difficulties greater than 1.5, and three items (i.e., Item 1, 7, and 9) took on negative values, suggesting that they were relatively easy. The other items were generally of medium difficulty.

Item and test information were also displayed to provide a more

Table 4
Item Difficulty of the Growing Pattern Test Items at T1 and T2.

Item	T1 δ	T1 s_e	T2 δ	T2 s_e
1	-.95	.18	-1.58	.23
2	.69	.16	.33	.16
3	1.79	.26	.30	.16
4	2.06	.29	.68	.17
5	1.26	.20	.12	.16
6	2.05	.31	.85	.18
7	.71	.17	-.01	.16
8	2.08	.31	1.17	.20
9	.89	.18	-.11	.16
10	2.49	.39	1.33	.21

Note. T1 $N = 124$. T2 $N = 120$. δ : item difficulty. s_e : standard error.

intuitive understanding of the precision of the test (see Fig. 2 and Fig. 3, respectively). In both figures, the horizontal axis represents children's ability in logit units (i.e., log odds units) while the vertical axis represents the amount of information the test provides at each ability level. Overall, the item and test information functions for the two subscales at T1 and T2 were bell-shaped. At T1, the information curves were negatively skewed, while at T2, skewness was negligible. For items, information peaks at about .80 at T1 and .60 at T2, which corresponds to an ability of approximately 0 to 3 and -2 to 2 on the logit scale, respectively. For the test as a whole, information peaks at about 3 for the two subscales at T1 and slightly more than 5 for the entire growing pattern test at T2, which corresponds to the ability ranges of 0 to 3 and -2 to 2 on the logit scale, respectively.

The finding that the two subscales at T1 showed the highest information when children's ability was in the range of 0 to 3 suggests that these scales were relatively difficult and appropriate for high-ability children. Outside of this range, the tests decreased sharply in the amount of information they could provide. This stands in contrast to the results at T2, where high information for children in the range of -2 to 2 supports the use of the growing patterns test for children of both low and high ability levels. Taken together, these findings suggest that while the growing patterns test was relatively difficult at T1, it became less difficult for these children at T2 and was an especially sensitive test for measuring their growing pattern understanding at this time point.

3.3. Sex DIF analysis

Prior to the formal analyses, a standard one-item-at-a-time anchor purification procedure was followed. None of the items exhibited DIF ($p > .05$). The sum score of individual items was then used as the matching criterion. The results of DIF analyses at T1 and T2 were reported in Table 5. The results of DIF analyses suggest that the items of the growing pattern test were free of sex bias and therefore fair to both girls and boys. Specifically, at T1, none of the 10 items exhibited statistically significant DIF across sex, as can be seen by the small M-H

Table 3
Infit and Outfit Statistics of the Growing Pattern Test Items at T1 and T2.

Item	T1 Infit		T1 Outfit		T2 Infit		T2 Outfit	
	MNSQ	ZSTD	MNSQ	ZSTD	MNSQ	ZSTD	MNSQ	ZSTD
1	.43	-4.74	.3	-1.57	.90	-.44	1.32	.64
2	.75	-2.05	.55	-1.44	1.02	.25	.92	-.2
3	.71	-1.39	.31	-.56	.96	-.33	.77	-.84
4	.96	-.05	.43	-.05	.95	-.32	.74	-.69
5	.75	-1.72	.37	-1.13	1.02	.21	.85	-.56
6	.63	-1.51	.27	-.91	.96	-.24	.75	-.52
7	.67	-2.59	.57	-2.16	.81	-1.85	.69	-1.38
8	1.00	.10	.62	-.19	.93	-.4	.80	-.21
9	.68	-2.57	.4	-2.75	.66	-3.51	.57	-2.04
10	.97	.04	.32	-.28	.86	-.77	.65	-.43
Item reliability: .98/.94*					Item reliability: .97			
Item separation: 7.61/4.00*					Item separation: 5.75			

Note. * Item reliability and separation were calculated separately for the increasing and decreasing patterns dimensions at T1. MNSQ: mean-square fit statistics, ZSTD: standardized fit statistic.

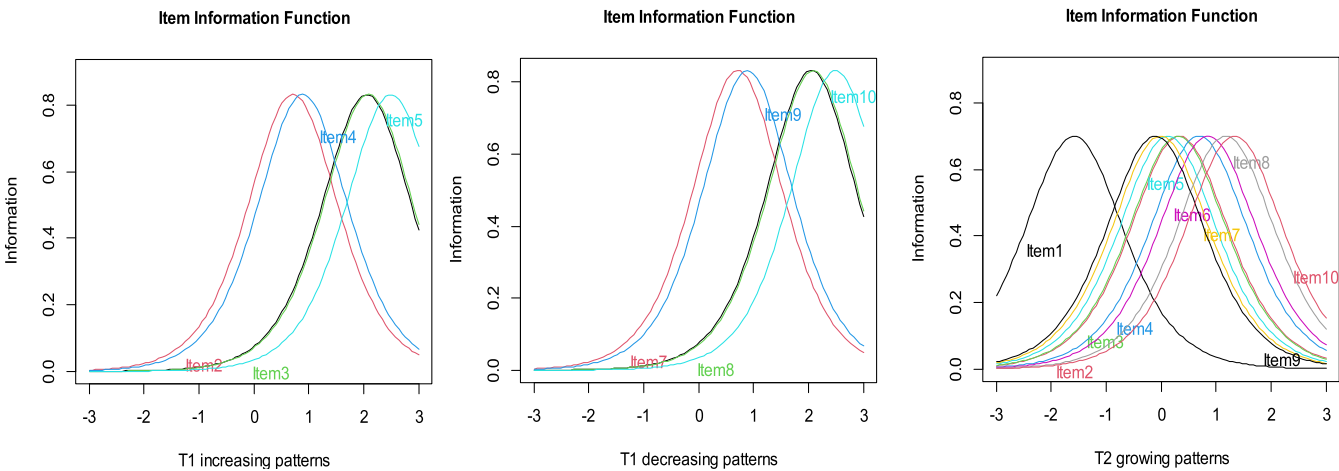


Fig. 2. Item Information Function of the Growing Pattern Test Items at T1 and T2.

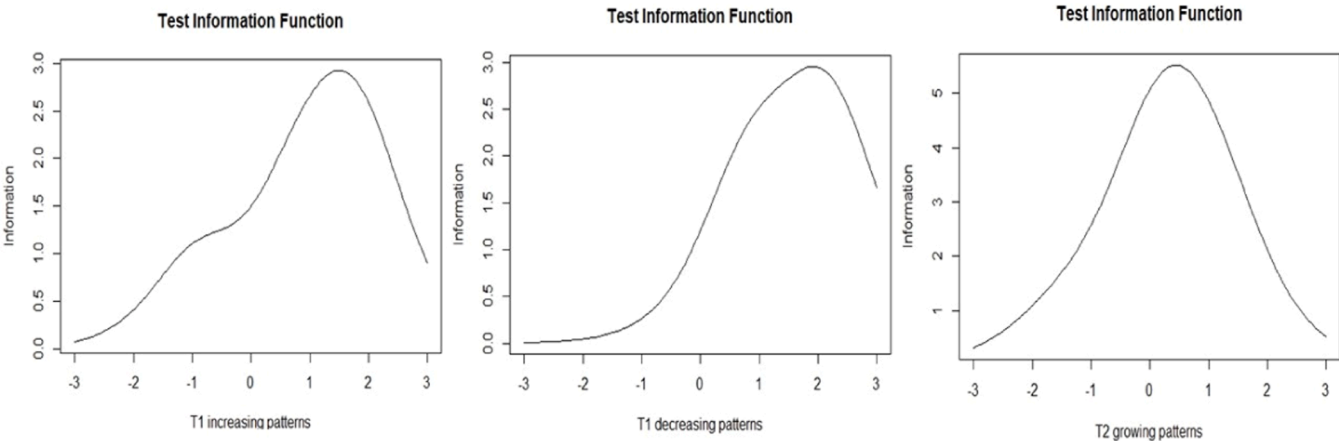


Fig. 3. Test Information Function of the Growing Pattern Test at T1 and T2.

Table 5
Identification of DIF Items in the Growing Pattern Test.

Item	T1			T2		
	M-H statistic	<i>p</i>	adjust. <i>p</i>	M-H statistic	<i>p</i>	adjust. <i>p</i>
1	.01	.91	1.00	.00	.94	1.00
2	.09	.77	1.00	.03	.85	1.00
3	.05	.82	1.00	.11	.74	1.00
4	.05	.82	1.00	.57	.45	1.00
5	1.44	.23	1.00	.23	.63	1.00
6	.02	.89	1.00	.13	.72	1.00
7	.20	.66	1.00	.55	.46	1.00
8	.02	.89	1.00	.60	.44	1.00
9	.01	.92	1.00	.01	.93	1.00
10	.00	.95	1.00	.32	.57	1.00

Note. T1 *N* = 124. T2 *N* = 120. adjust. *p*: adjusted *p* values were based on the Holm method.

statistics, large *p* values, and adjusted *p* values. Similarly, at T2, all items had nonsignificant results according to the adjusted *p* values, suggesting that they were free of DIF.

3.4. Convergent and discriminant validity

At T1, convergent and discriminant validity were established by correlating children’s scores on the increasing and decreasing pattern subscales and their performance on a repeating pattern task and an English expressive vocabulary task, respectively. A significantly positive and moderate correlation was found between repeating patterns and

increasing patterns ($r = .40, p < .001$) as well as between repeating patterns and decreasing patterns ($r = .38, p < .001$). In contrast, only a null and weak relationship was found between the two subscales and English expressive vocabulary (increasing patterns: $r = .19, p = .03$; decreasing patterns: $r = .07, p = .41$).

At T2, the same procedures for validation were followed except that the entire growing pattern scale was used rather than the two subscales. Similarly, a significantly positive and moderate correlation was found between repeating patterns and growing patterns ($r = .41, p < .001$), whereas only a weak to moderate correlation was found between growing patterns and English expressive vocabulary ($r = .29, p = .001$).

Together, these findings supported the convergent and discriminant validity of the growing patterns test at T1 and T2.

4. Discussion

The present study developed and validated a measure of growing pattern understanding for preschool children. The CFA results showed that growing pattern understanding consisted of two distinct albeit highly correlated components at PS2 (T1) but comprised only one factor at PS3 (T2), with adequate reliability at both time points. Rasch analysis of individual items suggested that all the items had acceptable difficulty, although most of them were relatively difficult at PS2 (T1). Convergent and discriminant validity were supported such that moderately positive correlations were found between growing and repeating patterns, while null and weak relationships were found with English expressive

vocabulary. These findings suggest that the growing pattern task is appropriate for use with five- and six-year-old children in Hong Kong.

4.1. The structure of growing pattern understanding

Existing studies mostly used a composite score to index children's growing pattern understanding, based on the untested assumption that this construct is unidimensional. However, as this study revealed, a one-factor model was insufficient to explain growing pattern understanding for children at PS2 (T1). Instead, increasing and decreasing patterns represented different components at PS2 (T1). This finding points to the possibility that for children aged five years, the type of growing patterns (i.e., increasing and decreasing patterns) may not be simply dismissed as a nuisance feature of task design. Instead, it could carry substantive meaning for the interpretation of children's scores on growing pattern tasks. To the extent that this is indeed the case, unidimensional scoring may mask different aspects of children's understanding of growing patterns and pose a threat to content and construct validity. This finding is especially pertinent as interest in growing patterns continues to rise and as more early childhood researchers begin to investigate preschoolers' knowledge in this domain beyond repeating patterns (Borriello et al., 2022; Pasnak et al., 2019; Wijns et al., 2019; Wijns et al., 2021). It is important that researchers dealing with this age group heed the psychometric evidence presented in this paper and investigate whether unidimensionality could be reasonably supported before they decide to use a single score to index children's understanding of growing patterns.

As no previous studies have presented psychometric evidence on growing patterns, we present our hypotheses regarding the division of growing patterns into two separate components at PS2 (T1). This could reflect the different cognitive demands in processing increasing and decreasing patterns. For example, a parallel can be drawn between increasing patterns and addition and between decreasing patterns and subtraction. The process of solving these pattern problems relies on children's understanding of addition and subtraction principles since children need to understand how many objects to add or remove to correctly solve the problems. There is broad agreement that subtraction problems are more difficult than addition problems (e.g., Kamii et al., 2001; Shinsky et al., 2009). In a similar vein, decreasing patterns may be more difficult than increasing patterns for these children. Indeed, the Rasch difficulty estimates in this study did suggest that increasing pattern items were relatively easier, as did the paired samples t-test result. Notably, while a two-factor model fit better at PS2 (T1), the two components were highly correlated ($r = .77$), suggesting that there was significant overlap. By PS3 (T2), when children became more competent in solving the growing patterns, the two components were statistically indistinguishable, and a one-factor model was sufficient for describing the data. This may reflect the possibility that at PS3 (T2), children were able to understand the abstract rule underlying growing patterns and the type of patterns (i.e., increasing and decreasing patterns) may no longer be relevant to children's substantive response processes.

Theoretically, if growing patterns for PS2 (T1) children indeed consist of two separate components, it begs the question as to whether these components have different predictive power for the same criterion variables or could predict outcomes in different domains. Future research can look into these questions using both types of growing patterns with children of diverse age ranges. From an assessment perspective, the findings of this study suggest that for younger children such as those approximately five years old, both increasing and decreasing types of patterns may need to be included. And interpretations of test scores should be conducted separately for the two dimensions. Yet for older children (i.e., children aged six), including increasing or decreasing patterns only when assessing growing patterns does not affect substantive interpretations. For potential intervention studies for young children, the findings here may also serve as a road map for what aspects of growing patterns to target in intervention designs.

4.2. Preschoolers' performance on the growing pattern task

This study corroborates previous findings that growing patterns might be difficult for four- to five-year-old children (e.g., Papic et al., 2011; Warren, 2005). In particular, Rasch analysis showed that most of the test items assessing children's understanding of growing patterns were difficult at PS2 (T1) when children were approximately five years old. However, at PS3 (T2), when children were about six years old, the item difficulties were well aligned with their abilities, suggesting that the test was a sensitive measure of children's understanding of growing patterns.

This study also adds to the current understanding of pattern understanding by revealing that increasing and decreasing patterns differ in their difficulty. At PS2 (T1), when growing pattern understanding could be decomposed into two separate components, paired-samples t-test showed that children performed significantly better on increasing patterns than on decreasing patterns (see *CFA analysis* section). Rasch difficulty estimates also showed that the items assessing decreasing patterns were generally more difficult than those for increasing patterns. Importantly, this pattern was maintained across time points. For task designers, a takeaway message may be that when designing tasks for assessing the growing pattern understanding of young children, both types of patterns can be included regardless of the factor structure of growing pattern understanding. In the case of a two-factor structure, including decreasing patterns can ensure construct validity. When only one factor is sufficient, incorporating decreasing patterns is not necessary but can still be useful as it can increase the overall difficulty of the task. For longitudinal projects in particular, where children need to be assessed repeatedly, including both types of patterns can expand the difficulty range of the test items and make them better-suited for children of diverse ability levels, thus ensuring adequate variability in the collected data.

The finding that most items provided adequate precision at PS3 (T2) also provides evidence against the claim that growing pattern understanding should be reserved for older children (Pasnak et al., 2019). In fact, many PS3 children (about six years old) in this study could solve growing pattern problems confidently regardless of pattern type, as evidenced by the alignment of item difficulties and children's abilities. Thus, this study supports previous findings that some children may already be capable of performing growing pattern tasks before they enter primary school (e.g., Borriello et al., 2022; Papic et al., 2011; Wijns et al., 2021). Therefore, assessing preschoolers' growing pattern understanding is feasible. Future studies should feel confident in utilizing our validated instrument for assessing growing pattern understanding for preschoolers who are approximately six years old.

4.3. Convergent and discriminant validity

In support of convergent validity, positive correlations were found between growing patterns and repeating patterns. The correlations were statistically significant but moderate in size, suggesting that they were distinct but related constructs. This is consistent with previous studies, which have found that growing patterns and repeating patterns are interrelated, as these two constructs assess different types of pattern understanding (e.g., Wijns et al., 2019; Borriello et al., 2022). Longitudinal studies have also shown that repeating and growing patterns are reciprocally related to each other from age four to age five (Wijns et al., 2021). This reciprocal relationship persists even after individual differences in prior patterning performance, spatial ability, and numerical skills have been taken into account. Given the existing evidence, the positive correlation observed in this study between the two constructs is expected.

For discriminant validity, we correlated growing patterns with English expressive vocabulary. Previous studies have shown that patterning skills may be related to some language skills such as early reading (Burgoyne et al., 2017; Kidd et al., 2014). However, this relationship is very tentative, and the observed relations tend to be much weaker compared with those between patterning and arithmetic skills

(Burgoyne et al., 2017). Therefore, we hypothesized that growing pattern understanding would be weakly correlated with or have a null relationship with English expressive vocabulary. These hypotheses were supported at both time points in our study. One explanation for the observed positive correlation is that pattern understanding, particularly the ability to abstract patterns and make predictions, might represent an early form of relational reasoning (Fyfe et al., 2015). To successfully complete a growing pattern task, such as extending a growing pattern ABABBABBB, children need to attend simultaneously to two key relations: one element keeps constant (i.e., A...A...A), and another element systematically increases (i.e., B...BB...BBB). Successful performance on this task item requires children to integrate the two relations to arrive at a correct response (i.e., AB BBB). This involves some degree of relational reasoning, which is foundational for children's cognitive development and may underlie children's acquisition of language-related skills (Pasnak et al., 2015). However, with very limited empirical evidence, our explanation remains hypothetical.

4.4. Contributions, limitations, and future research

To our knowledge, the current study is the first to investigate the factorial structure of growing pattern understanding among preschool children. This contributes to the literature by complementing the limited validation work in pattern understanding in general and growing pattern understanding in particular (Burgoyne et al., 2017). The findings of this research offer valuable insights into the dimensionality of growing patterns at different stages in the preschool period. Armed with this information, researchers and task designers can design their research projects and relevant growing pattern tasks accordingly so that they accurately reflect the nature of growing patterns. For potential interventions, this study may also provide some inspiration for what aspects of growing pattern understanding to target. Finally, preschool teachers can also incorporate the findings of this study into their educational practice by considering whether to include increasing or decreasing patterns or both in their daily instruction.

These contributions notwithstanding, a few limitations must also be acknowledged. First, despite the promising psychometric evidence presented in this study, the sample size used for validation is relatively small. With a small sample size, it is likely to capitalize on chance variation in the data and have low statistical power. While the model-fit results seem credible given the chi-square test and the other fit statistics, a larger sample size could have given us greater confidence in our findings. Future research could recruit more participants and conduct more systematic validation using a larger sample to verify our results. Second, only two assessment waves were included in our analysis and the task showed different factor structures at the two time points. Without more waves of measurement, we could not determine when dimensionality starts to change. Also, had we assessed children in primary school and found that growing pattern understanding continues to be unidimensional at this stage, we might have been able to conduct longitudinal measurement invariance to see whether its psychometric properties remain stable over time. These questions may be addressed in future studies. Third, when assessing growing pattern understanding, we only used a limited set of activities and task formats. This precludes us from making any inference about the impact of these task design features on children's performance. Future research could incorporate diverse task features into their design and use more sophisticated procedures, such as the multitrait-multimethod matrix, to disentangle the effects of task attributes and children's actual abilities. Finally, an additional limitation relates to the generalizability of our findings. As all our participants were Cantonese speakers recruited from Hong Kong, the findings could not be easily generalized to other cultural contexts. Also, our participants mostly came from middle- to upper-class families, the extent to which the findings could be generalized to lower-class families is also limited. Future research could greatly benefit from using larger and more diverse samples that cut across various cultural and national boundaries and span a broader spectrum of socioeconomic backgrounds.

5. Conclusion

This study developed and validated a measure of growing pattern understanding in a group of Hong Kong preschoolers who were followed from PS2 (T1) to PS3 (T2) over a one-year period. CFA analyses showed decreasing dimensionality of growing pattern understanding, with the construct being two-dimensional at PS2 and one-dimensional at PS3. Rasch analysis showed that all the items were within an acceptable range of difficulty. No differential item functioning across sex was found for the test items. Convergent and discriminant validity were supported by expected correlations between growing pattern understanding and repeating patterns and English expressive vocabulary. The findings of this study suggest that the task can be used as a reliable and valid instrument for assessing growing pattern understanding of five- and six-year-old preschool children.

CRedit authorship contribution statement

Xueliang Chen: Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing, Visualization. **Yawei Yang:** Conceptualization, Writing – original draft, Writing – review & editing. **Xiao Zhang:** Conceptualization, Data curation, Funding acquisition, Investigation, Project administration, Resources, Software, Supervision, Validation, Writing – review & editing.

Declaration of Competing Interest

None

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ecresq.2023.11.006](https://doi.org/10.1016/j.ecresq.2023.11.006).

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